Flexible quadrilateral mesh of isogonal type in arbitrary size

YANG LIU* and JINSAN CHENG✉*, Academy of Mathematics and System Science, CAS, China

CCS Concepts: • **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

ACM Reference Format:
Yang Liu and Jinsan Cheng. 2018. Flexible quadrilateral mesh of isogonal type in arbitrary size. *ACM Trans. Graph.* 37, 4, Article 111 (August 2018), 2 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Examples and visualization

For $i = 1, 2, 3, 4$, the variation of $x_i := \tan\left(\frac{\alpha_i}{2}\right)$ obey the so-called Bricard equation.

$$a_i x_i^2 x_{i+1}^2 + b_i x_i^2 + c_i x_{i+1}^2 + x_i x_{i+1} + e_i = 0, \tag{1}$$

where $x_5 := x_1$ and

$$\begin{cases} A_i := \cos(\mu_i') - \cos(\lambda_i' + \gamma_i' + \delta_i'), \\ B_i := \cos(\mu_i') - \cos(\lambda_i' + \gamma_i' - \delta_i'), \\ C_i := \cos(\mu_i') - \cos(\lambda_i' - \gamma_i' + \delta_i'), \\ D_i := 4\sin(\gamma_i')\sin(\delta_i') \neq 0, \\ E_i := \cos(\mu_i') - \cos(\lambda_i' - \gamma_i' - \delta_i'), \\ (a_i, b_i, c_i, e_i) := \left(\frac{A_i}{D_i}, \frac{B_i}{D_i}, \frac{C_i}{D_i}, \frac{E_i}{D_i}\right). \end{cases}$$

We only consider isogonal meshes in which

$$\forall i,\ \lambda_i' = \mu_i', \gamma_i' = \delta_i',$$

so that $b_i = c_i = 0$ and Eq. (1) becomes

$$(x_i y_i - k_i)(x_i y_i - K_i) = 0$$

where $\{k_i, K_i\} = \left\{\frac{-1 \pm \sqrt{1 - 4a_i e_i}}{2a_i}\right\} = \left\{-\frac{\sin(\gamma_i') \pm \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')}\right\}$.

Theorem 1. *A 3×3 isogonal mesh in Fig. 1 is flexible if and only if*

$$\forall i \in \{1, 2, 3, 4\}, \exists k_i \in \left\{-\frac{\sin(\gamma_i') \pm \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')}\right\} \ s.t.\ k_1 k_3 = k_2 k_4.$$

Example 2 (General case). Given an isogonal p-mesh as in Fig. 1. For $i \in \{1, 2, 3\}$, let $k_i = -\frac{\sin(\gamma_i') - \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')}$ and set $t = k_1 k_3 / k_2$. Then, the mesh is flexible if

$$\gamma_4' = \arctan\left(\frac{(1 + t^2)\cos(\lambda_4') + 2t}{(1 - t^2)\sin(\lambda_4')}\right).$$

NOTE: The range of arctan is taken as $(0, \pi)$ and $\arctan(\infty) = \frac{\pi}{2}$.

Authors' Contact Information: Yang Liu, yangrio@amss.ac.cn; Jinsan Cheng, jcheng@amss.ac.cn, Academy of Mathematics and System Science, CAS, Beijing, China.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 1557-7368/2018/8-ART111
https://doi.org/XXXXXXX.XXXXXXX

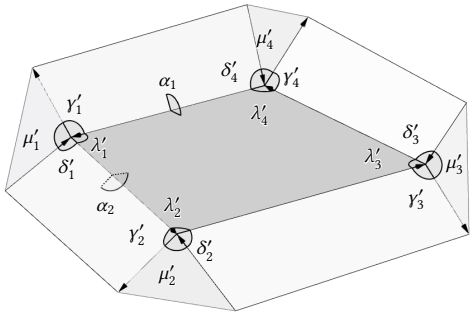


Fig. 1. Left: Sketch of a (trimmed) 3 by 3 p-mesh with fixed interior angles $\lambda_i', \gamma_i', \mu_i', \delta_i' \in (0, \pi)$ of planar faces. Flexible angles α_i are dihedral angles between the central face and the attached faces. For i odd (even), α_i is measured above (beneath) the central face.

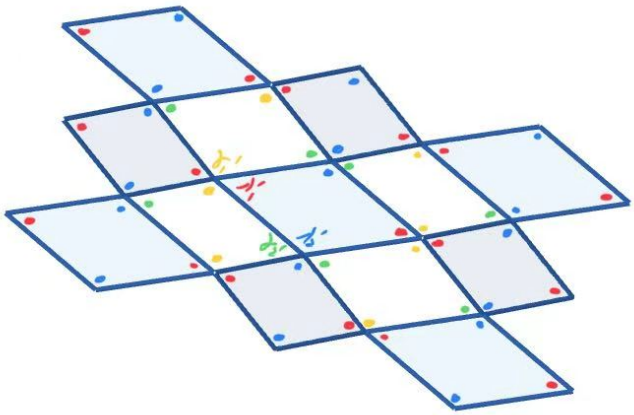


Fig. 2. Angles in the same color are identical; $\lambda_2' = \pi - \lambda_1', \gamma_2' = \pi - \gamma_1'$.

Example 3 (Parallelogram case for p-mesh). Choose $\lambda_1', \gamma_1' \in (0, \pi)$, and, as in Fig. 2, set

$$\lambda_2' = \pi - \lambda_1',\ \lambda_3' = \lambda_1',\ \lambda_4' = \lambda_2',\ \gamma_2' = \pi - \gamma_1',\ \gamma_3' = \gamma_1',\ \gamma_4' = \gamma_2'.$$

Moreover, for $i \in \{1, 2, 3, 4\}$, set

$$k_i = -\frac{\sin(\gamma_i') - \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')},\ K_i = -\frac{\sin(\gamma_i') + \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')}.$$

One can check that

$$k_1 k_3 = k_2 k_4,\ K_1 K_3 = K_2 K_4,\ k_1 K_3 = K_1 k_3 = k_2 K_4 = K_2 k_4.$$

https://www.geogebra.org/calculator/hsd9hkzw

Example 4 (Congruent case for sq-mesh). Choose $\lambda_1', \lambda_2' \in (0, \pi)$ such that $\lambda_1' + \lambda_2' < \pi$, hence non-planar mesh quads. As in Fig. 3,

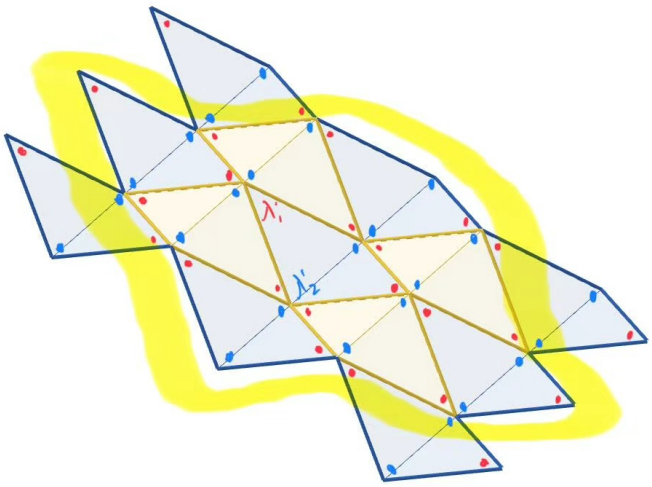


Fig. 3. Angles in the same color are identical; $\lambda_1' + \lambda_2' < \pi$.

set

$$\lambda_3' = \lambda_1',\ \lambda_4' = \lambda_2',\ \gamma_1' = \lambda_2',\ \gamma_2' = \lambda_1',\ \gamma_3' = \gamma_1',\ \gamma_4' = \gamma_2'.$$

Moreover, for $i \in \{1, 2, 3, 4\}$, set

$$k_i = -\frac{\sin(\gamma_i') - \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')},\ K_i = -\frac{\sin(\gamma_i') + \sin(\lambda_i')}{\sin(\lambda_i' + \gamma_i')}.$$

One can check that

$$k_1 k_3 = k_2 k_4,\ K_1 K_3 = K_2 K_4.$$

https://www.geogebra.org/calculator/mj826xqp

References

A Research Methods

A.1 Part One

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi malesuada, quam in pulvinar varius, metus nunc fermentum urna, id sollicitudin purus odio sit amet enim. Aliquam ullamcorper eu ipsum vel mollis. Curabitur quis dictum nisl. Phasellus vel semper risus, et lacinia dolor. Integer ultricies commodo sem nec semper.

A.2 Part Two

Etiam commodo feugiat nisl pulvinar pellentesque. Etiam auctor sodales ligula, non varius nibh pulvinar semper. Suspendisse nec lectus non ipsum convallis congue hendrerit vitae sapien. Donec at laoreet eros. Vivamus non purus placerat, scelerisque diam eu, cursus ante. Etiam aliquam tortor auctor efficitur mattis.

B Online Resources

Nam id fermentum dui. Suspendisse sagittis tortor a nulla mollis, in pulvinar ex pretium. Sed interdum orci quis metus euismod, et sagittis enim maximus. Vestibulum gravida massa ut felis suscipit congue. Quisque mattis elit a risus ultrices commodo venenatis eget dui. Etiam sagittis eleifend elementum.

Nam interdum magna at lectus dignissim, ac dignissim lorem rhoncus. Maecenas eu arcu ac neque placerat aliquam. Nunc pulvinar massa et mattis lacinia.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009